

On the Common Pitfalls of Designing and Communicating Within-Subjects Experiments in HCI

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Abstract

Well-designed experiments are essential for drawing valid statistical conclusions. In studies with limited resources (e.g., access to human participants), researchers often assign multiple conditions to the same participant (i.e., within-subjects experiments). Although such designs can increase statistical power, dependencies across trials within a participant may threaten the validity of the experiment. Unfortunately, domain-specific assumptions about these dependencies are often left implicit when conducting, analyzing, and communicating results from within-subjects experiments. We show that some common within-subjects experiments in the HCI community make assumptions that may not actually hold and provide a formal representation for precisely encoding these assumptions. We hope these results and examples will motivate changes to how we as a community reason about, design, and communicate experiments.

CCS Concepts

• Human-centered computing → Laboratory experiments.

Keywords

Experimental design, Within-subjects, Causality

ACM Reference Format:

London Bielicke, Emery D. Berger, Adam Chlipala, and Eunice Jun. 2026. On the Common Pitfalls of Designing and Communicating Within-Subjects Experiments in HCI. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799010>

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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3799010>

1 Introduction

Well-designed experiments are essential for scientific validity and reproducibility. Researchers design experiments to establish causal relationships, such as determining the effect of a new interface on a user's ability to complete a task. When experiments are poorly designed, researchers may mistakenly attribute the effect of the experimental setup itself to the interface, leading to incorrect causal conclusions.

Unfortunately, ideal experimental conditions are often infeasible due to practical considerations, such as constraints on access to participants, time, or other limited resources. Instead, within-subjects designs are a common approach to increase statistical power in small-n experiments by having each participant serve as their own control. They are widely used in fields such as psychology [4], biomedicine [10], and HCI [8]. Concretely, more than 60.8% of papers that use experimental design keywords in CHI 2023–2025 proceedings also use keywords relating to within-subjects designs.¹ In these experiments, researchers randomly assign sequences of conditions (e.g., baseline → interface) to participants rather than individual conditions (e.g., baseline).

The validity of within-subjects experiments depends on domain-specific assumptions about *order effects*. These effects occur when earlier trials influence outcomes in subsequent trials. For example, consider an experiment in virtual reality (VR) in which “cybersickness” from the VR task persists and impairs performance in subsequent trials. If researchers fail to distinguish the direct effect of the baseline condition from the effect of lingering cybersickness, they may overestimate or underestimate the causal effect of the VR interface. Such effects can also arise due to skill transfer. For example, in an experiment comparing two statistical modeling packages, a package with more extensive documentation may leave participants with more knowledge of statistical modeling choices, improving their performance with the baseline system in later trials.

¹We first searched CHI 2023–2025 proceedings for papers with experimental design keywords. Among these, we identified those mentioning within-subjects designs using terms such as “repeated measures,” “within-subjects,” “within-participants,” “crossover design,” “Latin square,” and “counterbalancing.”

Although common within-subjects designs in HCI account for some order effects, they do not account for the lingering effects of cybersickness. For example, Latin squares ensure that each condition appears equally in each trial position, which balances systematic changes over time that affect all conditions equally [8]. However, because each condition never precedes itself in a sequence, separating the lingering effects of a previous condition from the effects of the current condition on participant outcomes remains challenging [14]. The distinction is subtle, and conflating these different types of order effects can lead to mismatches between domain assumptions and design choices.

Despite their importance, assumptions about order effects are rarely made explicit in HCI papers, making it difficult to assess experimental design validity. Of the 793 papers that mention within-subjects designs in CHI 2023–2025 proceedings, only 97 (12.2%) address order effects at all.² Among these 97 papers, few specify which order effects they assume or which their designs address. For example, some papers use Latin squares “to mitigate the influence of order effects and immediate carry-over effects” [9]. Others claim they use them to “prevent any carryover effects” [5]. This language is imprecise or incorrect: Latin squares balance certain order effects across participants but do not eliminate carryover effects (Section 2). Moreover, even when a design accounts for carryover effects, it cannot prevent them from occurring.

To ensure proper experimental design and accurate interpretation of experimental findings, HCI researchers need a precise way to specify and distinguish between possible order effects. Existing frameworks, such as graphical causal models (GCMs), provide ways to represent and reason about causal assumptions. However, no prior work provides a systematic method for reasoning about how participants evolve over the course of an experiment and for deriving experimental designs that are compatible with these assumptions. To bridge this gap, we leverage *time-indexed causal graphs* to ground assumptions about order effects and guide design of within-subjects experiments. We view these causal graphs as a cognitive tool to scaffold scientists’ thinking and workflows towards more rigorous and reproducible experimental practices in HCI.

Our contributions include:

- (i) A framework for explicitly representing assumptions about order effects using time-indexed causal graphs
- (ii) Procedures to detect bias in within-subjects experimental designs under these assumptions.
- (iii) Practical guidance for selecting appropriate designs given researchers’ assumptions about order effects.

2 Threats to Validity of Within-Subjects Experiments

Order effects are the primary threat to validity in within-subjects experiments. They lead to systematic variation in participant outcomes due to the sequences of conditions they previously experience [2]. There are two types of order effects: period effects and

carryover effects. Distinguishing between them is essential for correct design and analysis of within-subjects experiments, as they introduce distinct biases but are easy to conflate in practice.

Period Effects. Period effects occur when the outcome of a trial varies systematically based on when it occurs (e.g., trial 1 vs. trial 2), regardless of which condition was previously assigned [15]. For example, participants may improve at creating a website in trial 2 simply because they gained experience creating websites in trial 1. Counterbalanced designs prevent these period effects from biasing causal-effect estimation by ensuring that each condition appears equally often in each trial position.

Carryover Effects. Carryover effects occur when the lingering effects of conditions assigned in previous trials influence outcomes in subsequent trials [15]. Unlike period effects, carryover depends on which specific condition participants experienced previously. For example, suppose a researcher is testing the effect of VR environments on embodiment. Feelings of cybersickness may persist after leaving the virtual environment, leading to decreased feelings of embodiment in the baseline condition.³

“Higher-order carryover” occurs when prior *sequences* of conditions influence the outcome of a trial [3]. Accounting for higher-order carryover, either through experimental design or statistical adjustment, quickly becomes a combinatorial problem, as the number of possible condition sequences grows rapidly with the number of conditions.

3 Motivating Example

We demonstrate the importance of communicating a-priori assumptions using a modified version of the within-subjects experiment in “I think I don’t feel sick: Exploring the Relationship Between Cognitive Demand and Cybersickness in Virtual Reality using fNIRS” published at CHI ’23 [13]. The researchers investigated the effect of visual motion on cybersickness using a Latin square design (Table 1). They did not explicitly state any assumptions about order effects.

Table 1: Sequences in the Latin square design used to study the effect of motion on cybersickness.

Sequence	Trial 1	Trial 2
1	Motion	No Motion
2	No Motion	Motion

During analysis, the authors found a significant motion-by-time (i.e., motion-by-period) effect, reporting that “time had a stronger effect in conditions in which the roller coaster was moving compared to the ones in which it was stationary” [13]. Carryover effects provide one possible explanation for the observed motion-by-period effect [15]. However, their Latin square design (Table 1) ensures participants never experience the same condition consecutively, making it impossible to distinguish carryover effects from period effects during analysis.

³This phenomenon has been observed in existing virtual-reality studies [1].

²We searched CHI 2023–2025 papers for keywords related to order effects, including “period effects,” “carryover effects,” “learning effects,” and “practice effects.” Period and carryover effects are two distinct types of order effects, defined in Section 2, while learning and practice effects are common examples.

Without clear assumptions about carryover effects, the experimental findings indicate either that (i) motion has a larger direct effect on cybersickness in later trials or (ii) cybersickness from the motion condition lingers and affects subsequent trials (a carryover effect). Had the authors explicitly considered carryover assumptions during design, they could have chosen an alternative design (discussed in greater detail in Section 4.4.2) that enables separate estimation of carryover effects [7]. If substantial carryover is present, the estimated effect of motion may be confounded with lingering effects from prior trials, leading to biased causal estimates [15].

4 Graphical Representation of Within-Subjects Designs

Graphical causal models provide a formal way to represent and reason about causal relationships [12]. However, existing work does not demonstrate how to apply these procedures to reason about carryover effects in HCI experiments.

Our key insight is that we can model within-subjects designs using one treatment, one outcome, and zero or more state nodes for each trial a participant experiences, creating a *time-indexed causal graph*. State nodes (Figure 1) include attributes (e.g., experience) or environmental factors (e.g., time of day).

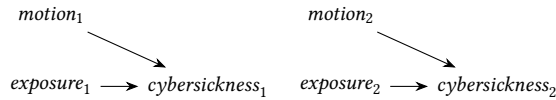


Figure 1: Base graph for a two-trial within-subjects design with prior exposure to VR systems ($exposure_i$) included as a state variable.

We build on prior work that formalizes two-trial within-subjects designs using GCMs [6], generalizing their approach to arbitrary within-subjects designs and providing guidance for constructing these graphs. The graphs enable researchers to (i) explicitly represent carryover assumptions, (ii) distinguish between period and carryover effects, and (iii) evaluate whether their experimental design produces unbiased estimates given these assumptions.

Below, we construct a graph based on the example described in Section 3. The graph makes assumed order effects explicit. By comparing two experimental designs, we show how different design choices can affect analysis when carryover is present.

4.1 Graph Construction

To start, we include two copies of motion and cybersickness in the graph. Within each trial, arrows from $motion_i$ to $cybersickness_i$ represent the causal relationship of interest (Figure 2a).

The key difference from standard causal graphs is that researchers can include edges between trials, representing carryover effects (Figure 2b). For example, $motion_1 \rightarrow cybersickness_2$ represents the assumption that the condition assigned in the first trial has a lingering effect on cybersickness in the second trial. Prior work uses this approach to represent unit interference in an experiment [11].

Although the researchers did not explicitly state this assumption, lingering effects of cybersickness are a well-documented issue in

VR studies [1] and may explain the motion-by-time effect observed in the analysis.

4.2 Detecting Carryover Effects

Carryover effects reflect changes between trials caused by previously administered treatments. They provide more precise explanations of changes in outcome over time but are more difficult to address by design.

We provide a general criterion for identifying carryover effects in within-subject GCMs, where T_i represents a treatment node (e.g., motion) at trial i , and Y_i represents an outcome node (e.g., cybersickness) at trial i . Formally, carryover effects exist if, for any treatment T_i , there exists a d-connected path [12] (i.e., an open path indicating direct or indirect association) from T_i to Y_j for $i < j$ that does not pass through treatment T_j . In other words, a carryover effect exists when an earlier treatment can still influence a later outcome through some chain of causal dependencies. In Figure 2b, the path from $motion_1 \rightarrow cybersickness_2$ encodes an assumption about carryover effects.

4.3 Detecting Period Effects

Period effects reflect systematic changes in outcomes between trials, regardless of the assigned condition(s). Because time (i.e., trial number) is encoded in causal graphs through node indexing rather than as an explicit variable, period effects manifest as distributional differences across outcome variables (e.g., $cybersickness_1$ and $cybersickness_2$), rather than as edges in the causal graph.

4.4 Checking for Carryover-Induced Bias

We can use our graphical representation of carryover assumptions to determine whether the experimental design produces unbiased estimates.

Backdoor paths and confounding. Causal effects exist when there is no confounding between treatment and outcome. In causal graphs, confounding is indicated by open “backdoor paths”: paths from treatment to outcome that enter the treatment node with an arrow pointing in [12]. To obtain unbiased estimates, all backdoor paths (i.e., non-causal relationships between a causal and a treatment) must be blocked.

Design-induced edges. In within-subjects experiments, the design itself can create dependencies between treatment assignments across trials. When the treatment assigned in one trial provides information about treatments in other trials, we add edges between treatment nodes in the graph.

4.4.1 Example 1: Latin square. We show how the graph can indicate potential threats to validity in the experiment described in Section 3 under carryover assumptions.

Existing work demonstrates how the sequences in a Latin square design introduce edges in a causal graph due to dependencies between condition assignments across trials [6]. In our example, each participant is assigned one of two sequences: motion $\rightarrow$ no motion or no motion $\rightarrow$ motion. Thus, if participant 1 receives the motion condition in the first trial, they will receive no motion in the second trial. We represent this dependency in the causal graph by connecting $motion_1$ and $motion_2$ (Figure 2c).

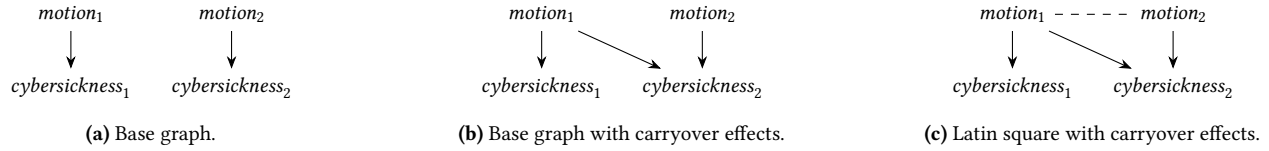


Figure 2: Graphical causal models (GCMs) for a two-trial within-subjects design.

Using existing causal algorithms, we can determine that the causal effect of motion on cybersickness is biased under carryover assumptions due to the backdoor path $motion_2 \leftarrow motion_1 \rightarrow cybersickness_2$.

4.4.2 Example 2: Sequential Randomization. We adapt the experimental design from Section 3 to show how design choices affect bias and interpretability in within-subjects experiments.

In this design, participants receive one of four possible sequences: motion $\rightarrow$ no motion, no motion $\rightarrow$ motion, motion $\rightarrow$ motion, or no motion $\rightarrow$ no motion. Because the motion condition is randomized across trials, there is no dependency between $motion_1$ and $motion_2$, so no edge connects them in the causal graph (Figure 2b).

Since there is no backdoor path from $motion_2$ to $cybersickness_2$ in the constructed graph, the causal effect of motion on cybersickness is unbiased, even under the assumed carryover effects.

4.4.3 General Criterion. We provide a general criterion for determining whether a design is compatible with assumptions about carryover effects.

For each treatment node T_i , check whether a backdoor path exists from T_i to the corresponding outcome node Y_i . If such a path exists, carryover effects may introduce bias in the estimated treatment effect.

5 Design Implications

Having shown how graphical models can make carryover assumptions explicit, we provide recommendations for researchers designing experiments based on these assumptions. We organize recommendations around three common assumptions about order effects, progressing from strongest to weakest assumptions.

5.1 Case 1: No Carryover Effects, No Period Effects

Assumption. We assume (i) no carryover exists and (ii) treatment outcomes are constant across periods. Under this assumption, there is no edge from $motion_1$ to $cybersickness_2$ in the causal graph (Figure 2a).

Use a within-subjects design where each participant experiences each condition an equal number of times. This provides maximum statistical power by having participants serve as their own controls. It is not necessary to counterbalance conditions because we assume that there are no period effects. For example, we could assign one sequence (motion $\rightarrow$ no motion) to all participants.

5.2 Case 2: No Carryover Effects, Period Effects May Exist

Assumption. We assume no carryover exists, but the period may affect observed outcomes. Under this assumption, there is no edge

from $motion_1$ to $cybersickness_2$ in the causal graph (Figure 2a), yet the expected value $cybersickness_1$ differs from the expected value of $cybersickness_2$.

Use a traditional within-subjects design (e.g., Latin square) with counterbalancing, which provides maximum statistical power by having participants serve as their own controls and ensures that conditions are evenly distributed across periods. Adjusting for period effects during analysis can further improve statistical power by accounting for systematic variation across trials.

5.3 Case 3: Both Carryover and Period Effects May Exist

Assumption. We assume both carryover effects and period effects may influence the outcome of trials. Under this assumption, there is an edge from $motion_1$ to $cybersickness_2$ in the causal graph (Figure 2b).

Since these assumptions are restrictive, we enumerate three options for HCI researchers to consider.

Option 1. When researchers can make assumptions about variables that influence the outcome (e.g., age), use a between-subjects design to avoid order effects and adjust for the variables during analysis to increase statistical power.

Option 2. When researchers are interested in measuring carryover effects, they can design a sequentially randomized experiment (Figure 2b). These designs may have lower statistical power than traditional within-subjects designs because not all participants experience all conditions (e.g., when they are assigned motion $\rightarrow$ no motion), meaning participants do not always serve as their own control [4]. Adjusting for the previous conditions can improve statistical power if carryover is present.

Option 3. When researchers want to check for carryover effects in traditional within-subjects experiments, they can measure relevant variables (e.g., cybersickness) before each trial. Significant differences in these variables between trials indicate the presence of carryover, which may bias the experimental findings. If carryover is likely, researchers can analyze the data from the first trial.

6 Conclusion

We leverage *time-indexed causal graphs* as a cognitive tool to formally specify order effects and reason about bias in within-subjects experiments. These graphs make explicit how experimental design choices influence causal conclusions and potential biases. Using these time-indexed causal graphs, we derive experimental design recommendations for HCI researchers based on their domain assumptions. Future work should focus on easing specification of these graphs. For example, a system could translate design specifications and researchers' domain assumptions into these causal

graphs. By mapping design choices to causal assumptions, the system may help automate the design of experiments under more fine-grained assumptions about order effects.

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